From GPS to Maps: Path Reconstruction Techniques for Understanding People Mobility with Low-Sampling-Rate Data

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***Abstract*—Urban mobility is a critical aspect of city planning and transportation management, with the Global Positioning System (GPS) playing an essential role in analyzing and understanding patterns of human movement. Particularly using smartphones, GPS provides highly detailed digital trace data, enabling in-depth analysis of human movement behavior. However, there are significant challenges when working with GPS datasets that have a low frequency, which often occurs due to storage capacity limitations or data transmission costs. This study explores methodologies for reconstructing movement paths from low-frequency GPS datasets. The process includes several key steps: data pre-processing involving cleaning and normalizing data; aligning GPS data with existing road networks using the nearest-neighbor KD-Tree algorithm; and accurate path reconstruction. In addition, the study develops a method for time interpolation estimation to determine movement times between the aligned coordinate points. The findings of this research indicate that with the proposed approach, movement paths can be reconstructed effectively and accurately, even from data with low sampling frequencies. This approach promises a wide potential application, ranging from transportation infrastructure planning to the development of more sophisticated navigation systems. With ongoing research, the integration of additional sensors and machine learning techniques is anticipated to further enhance the accuracy and utility of this methodology.**

***Index Terms*—GPS, people mobility, KD-Tree, map-matching, digital footprint, path reconstruction.**

I. Introduction

Global Positioning Systems (GPS) have undergone significant transformations in their use since being introduced several decades ago. Initially known as a navigation technology, GPS has evolved far beyond its original role. In today's digital era, where almost everyone has a smartphone, GPS not only functions as a navigation tool but also as a digital human footprint [1]. Each smartphone, with its GPS sensor, records location data that can provide insights into how and where a person moves.

With the ability to track individual movements in real-time, GPS technology, especially that found in smartphones, has become an important instrument in understanding human mobility [2]. By mapping individual movements, we can gain deep insights into travel patterns, social interactions, and many other aspects of daily life. Human mobility represents a form of dynamic in spatiotemporal phenomena, such as urban mobility, goods distribution, and pollutant diffusion, supporting findings [3] [4]. Studies on human mobility are very important in various fields, including urban planning and traffic management [5]. Due to its wide spatiotemporal coverage and low operational costs, cell phone data has been recognized as a primary data source for many studies and applications related to mobility [6].

However, in many applications, especially those related to urban planning and urban resilience, high-samplig-rate GPS data sampling may not be practical or too expensive. Instead, low-sampling-rate data [2] [7] [8] [9] often becomes the choice.

A. Challenges in Using GPS Data with Low Sampling

GPS data with a low sampling rate poses significant challenges in mobility analysis. Unlike high-sampling data, it may not capture the intricacies of a person's movements, complicating accurate path reconstruction. This necessitates specialized techniques to optimally utilize the data and derive valuable insights. Despite numerous studies, two notable gaps remain unaddressed: first, the use of graph-based approaches for path reconstruction to represent road networks effectively, and second, the lack of discussion on time calculation between GPS points.

B. Objectives and Main Contributions

In this context, the primary objective of this paper is to elucidate and evaluate path reconstruction techniques that can be utilized to comprehend people mobility from low-sampling-rate GPS data. The contributions of this paper include:

1) Introducing an innovative method for handling the analysis of low-sampling-rate GPS data.

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2) Implementing spatial data compression to optimize computational load.
3) Adopting advanced time interpolation techniques to determine timestamps for additional coordinates, ensuring accurate trajectories.

By understanding individual movement within the context of low-sampling-rate data, we can better assist policymakers and city planners in effectively addressing the needs of the populace [5], all while maintaining urban resilience amidst various challenges.

II. Related Works

In recent years, extensive research has been conducted on reconstructing travel paths from GPS data. With the rising popularity of GPS use, particularly through smartphones, the need for a more accurate understanding of individual travel patterns has become increasingly critical. Traditional techniques often rely on algorithms that identify key points in GPS data and connect these to reconstruct an individual's traveled path.

Map-matching is an essential process that aligns a collection of GPS data points with actual road networks [8] [11]. Various map-matching approaches are noted in the literature, including geometric, topological, and statistical methods. For instance, Hsueh et al. [2] proposed a map- matching algorithm known as STD-matching for low-sampling-rate GPS data, which incorporates spatial, temporal, and directional analyses to find the best match for GPS trajectories. Hashemi et al. [11] developed a map-matching technique using Artificial Neural Networks (ANN) to enhance the accuracy of GPS-based map-matching algorithms. Meanwhile, Sinthopvaragul et al. [8] introduced a map-matching method employing Machine Learning approaches on low sampling rate GPS data, comparing methods such as XGBoost, Decision Tree, KNN, MLP, and Random Forest. Du et al [12], offered an HMM-based approach named Fast Global Map-Matching (FGMM).

Other researchers, like Luo et al. [12] , have also utilized low sampling rate GPS data, applying the INC-RB algorithm which proved to be more efficient than the Hidden Markov Model (HMM). In the context of path reconstruction using cell phone data, research by Chen et al. [13] employed Mobile Positioning Data Passive (MPD Passive), selected for its ability to cover a broader population. Another case, Syarif et al. [14] uses MPD Passive data to estimate public transport destinations.

III. Data Acquisition

A. How GPS Generates Coordinate Data

The Global Positioning System (GPS) is a technology that has revolutionized our understanding of location and movement. Comprising a network of about 24 satellites orbiting the Earth, GPS provides accurate location information up to a few meters.

A GPS device is activated, it begins to search for signals from satellites within its range. To determine its position accurately, the device typically requires signals from at least four satellites. Upon receiving these signals, the device measures the time taken for the signals to reach it from each satellite, allowing it to calculate its distance from every satellite.

Knowing the distance from several satellites, the device then undergoes a process known as triangulation to determine its geographical position on Earth. The result of this triangulation is presented in the form of geographical coordinates, namely latitude and longitude. Some advanced GPS devices are also able to determine elevation or height relative to sea level.

Over time, by obtaining coordinates periodically, the GPS device can track the user's movement. This creates a mobility trail that can be recorded and analyzed. In the context of mobility data, each entry typically includes coordinates and a timestamp, providing a detailed view of how and when the movement occurred.

B. High Sampling Rate vs Low Sampling Rate

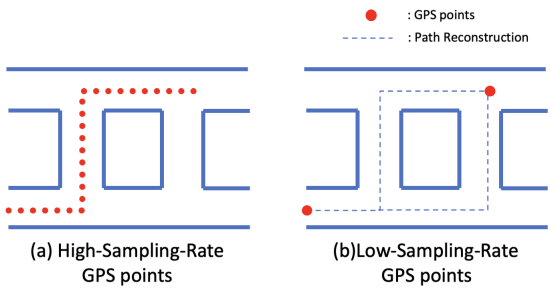

Fig. 1. (a) High-Sampling-Rate, (b) Low-Sampling-Rate.

High-Sampling-Rate GPS Points Fig. 1.(a) :

1) High-sampling-rate GPS points refer to GPS data that is collected at short, regular intervals. This results in a dense set of location points that can provide a very detailed representation of a user's path or movement over time.
2) For instance, if a GPS device is set to record location data every second or every few seconds, it would be considered high-sampling-rate.
3) The advantages of this approach include higher accuracy in mapping the actual path taken and the ability to detect minute changes in direction or speed.
4) However, the drawback is that it generates a large volume of data, which might be challenging to store, transmit, or process, especially in real-time scenarios or with limited computational resources.

Low-Sampling-Rate GPS Points Fig. 1.(b) :

1) Low-sampling-rate GPS points are the opposite; they refer to GPS data collected at longer, irregular intervals. This might result in a sparser set of location points, which might not capture every minor detail of a user's movement.
2) For example, if a GPS device records location data every few minutes or even longer, it would fall into this category.
3) The main advantage here is that it generates a smaller volume of data, making it easier to store, transmit, and process. This can be especially useful for applications where real-time processing isn't crucial or where bandwidth and storage are limited.
4) The challenge with low-sampling-rate data is that it might miss out on some intricate details of the movement, leading to potential inaccuracies when reconstructing paths or when analyzing movement patterns.

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C. Dataset

In this study, historical mobility data of individuals from November 2021 to June 2022 was utilized. The data comprises low-sampling-rate GPS points detailed in four columns, as described in Table I.

TABLE I
INFORMATION ON GPS DATASET

Information	Description
maid	Encrypted unique ID of the user
latitude	Estimated longitude coordinates of the device location.
longitude	Estimated latitude coordinates of the device location.
timestamp	The time (in UNIX form) at which the data was sampled.

TABLE II
RAW DATA OF GPS DATASET

maid	latitude	longitude	timestamp
b74fb520-8f92-9771-affc32c6af5d	-7.79426	110.36891	1638291600
b74fb520-8f92-9771-affc32c6af5d	-7.79424	110.36891	1638291604
b74fb520-8f92-9771-affc32c6af5d	-7.79424	110.36891	1638291607
8e8ab5b5-5bef-4e12-i460a97a13e1	-7.83484	110.32333	1638291601
8e8ab5b5-5bef-4e12-i460a97a13e1	-7.83481	110.32331	1638291610

Table II illustrates that 'maid' represents the unique ID of users. 'Latitude' and 'longitude' specify the geospatial location, while 'timestamp' indicates the time the data was recorded, using the Unix Timestamp format. The Unix Timestamp system tracks time by counting the number of seconds that have elapsed since the Unix "epoch," which is 00:00:00 UTC on January 1, 1970 (excluding leap seconds). In other words, it is a global timekeeping system used by operating systems and internet protocols to represent specific points in time. Prior to analysis, timestamps must be converted into the MM-DD-YY HH:MM:SS GMT+7 format.

D. Road Network Data

The road network data was obtained for free from OpenStreetMap in the form of a shapefile. Initially, the geometry of the road network data is in Linestring format. This data was then processed and transformed into a graph structure, as the fundamental nature of a graph is well-suited to represent road networks. Table III presents an example of the road network data for the Special Region of Yogyakarta.

TABLE III
ROAD NETWORK DATA

name	start x	start y	end x	end y
Kukup – Drini – Indrayanti	110.567130	-8.131536	110.566176	-8.131169
Kukup – Drini – Indrayanti	110.567130	-8.131536	110.568241	-8.131745
Segoroyoso – Sitimulyo	110.417259	-7.870120	110.416739	-7.871060
Segoroyoso – Sitimulyo	110.417259	-7.870120	110.418351	-7.870158

IV. METHODOLOGY

The process flow of path reconstruction is illustrated in Fig. 2. The process begins with raw GPS data, which undergoes preprocessing to address any points located outside the road network; these are pulled to the nearest road in a step known as Map-Matching. Following this, data compression is performed by eliminating immobility data points. The final steps are path reconstruction and time interpolation. During path reconstruction, coordinates from vertices are added as auxiliary points to form a trajectory that aligns with the road network. Time interpolation then provides an estimated time of arrival at specific coordinate points. Each of these processes is described in greater detail as follows:

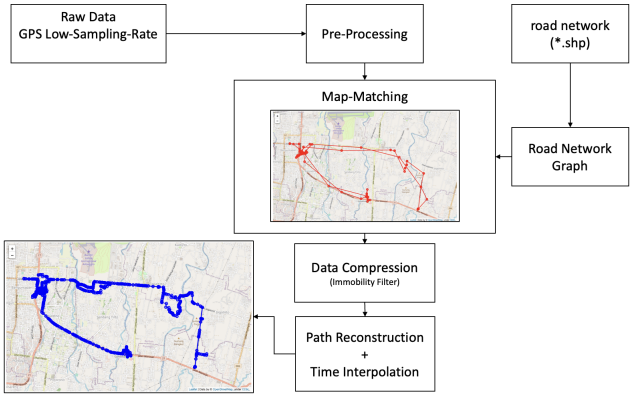

Fig. 2. Path Reconstruction Process Flow.

A. Preprocessing Data

Before analysis of GPS data can commence, certain preprocessing steps are essential to ensure the quality and integrity of the data. This process includes:

1) Selection of Relevant Columns: Not all columns in the original dataset may be pertinent to the analysis. Therefore, specific columns are selected based on their relevance to the analytical objectives.
2) Removal of Duplicate Data: Duplicate data can introduce bias in the analysis. Repetitive entries are identified and removed to guarantee the uniqueness of each data point.
3) Data Type Conversion: The timestamp columns often need to be converted into datetime data types for accurate analysis. This allows for time-based comparative analysis between data points.

B. Map Matching to Road Network

After preprocessing, the next step is to map the GPS data onto the road network.

1) Transforming Road Network Data into a Graph: The road network data, sourced from OpenStreetMap in shapefile format, was converted into a graph structure to model and analyze movement patterns and navigation. This transformation uses the properties of graphs for a more efficient and accurate representation of the road network. In the graph, vertices denote intersections or key points, and edges represent connecting road segments. This facilitates the analysis of shortest paths, traffic density, and movement patterns using existing graph algorithms and techniques.
2) Map Matching with KD-Tree: This method uses the KD-Tree data structure to efficiently map GPS data points to the nearest road network. It involves dividing the road network's two-dimensional space into distinct areas within the KD-Tree, each representing a road segment. The KD-Tree algorithm quickly searches and maps GPS points to the nearest road segment, reducing the need for comparisons compared to linear search. This ensures each GPS point accurately corresponds to a valid physical location on the road network, essential for traffic pattern analysis and routing in dense urban areas.

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3) Implementing an Immobility Filter: To reduce computational load, data is filtered to remove instances of immobility. This filter assesses the duration and distance between consecutive GPS points. If an object stays at the same coordinates for several time points, it's considered immobile. The study uses low-sampling-rate GPS data, but only data representing immobility is compressed. This means only data detected as stationary over consecutive periods is removed, as it doesn't provide significant meaning and burdens the computational process. The immobility filter works by discarding data points between the initial and final points of immobility, retaining only the start and end times of immobility. This helps determine the duration of an object's immobility at a point by examining the time difference between these two points. An illustration of the immobility filter process can be seen in Fig. 3. The method optimizes data for analyses like traffic flow or urban planning by focusing on movement. By analyzing the remaining data, more accurate patterns of mobility can be inferred. This filtering is especially useful in smart city applications where precise location tracking is crucial.

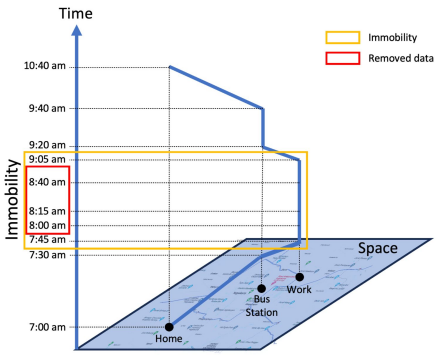

Fig. 3. Illustration of Immobility Filter.

C. Path Reconstruction

With the GPS data now mapped to the road network, we can reconstruct the paths traveled by individuals.

1) Shortest Path Search: Low-sampling-rate GPS data tends to produce points that may be far apart from each other, complicating the accurate reconstruction of paths. To address this, shortest path algorithms, such as A*, are adopted to estimate the most likely path taken between these points. The estimated path then serves as the basis for reconstruction.
2) Time-Ordered Reconstruction: Utilizing the timestamps from each GPS data point, the individual's path is reconstructed based on the chronological order, allowing us to track the movement of the individual over time.
3) Time Interpolation: The next challenge is that the coordinates added to form a path that aligns with the road network must estimate positions based on time. In path reconstruction from low-sampling-rate GPS data, a critical aspect is determining timestamps for each newly generated coordinate. This process is known as 'time interpolation'. First, we calculate the time difference, Δt, between two consecutive GPS points as in (1):

$$\Delta t = t_2 - t_1 \tag{1}$$

Given t_1 and t_2 as the timestamps from the two GPS points in question, we then estimate the duration between each new coordinate δt during the map-matching process, see (2). This is done by dividing the time difference,Δt, by $n+1$, where n is the number of new coordinates generated:

$$\delta t = \frac{\Delta t}{n+1} \tag{2}$$

Consequently, for each i new coordinate, the timestamp t_i can be calculated by (3) as follows. The variable i serves as an iteration index used in equation 3 to represent each newly generated coordinate during the map-matching process. ranging from 1 to n, i plays a crucial role in producing new coordinates and calculating the corresponding timestamps.:

$$t_i = t_i + (i \times \delta t) \tag{3}$$

This approach ensures that time interpolation assigns a consistent timestamp to each new coordinate, accurately reflecting the temporal movement of an individual along the reconstructed path.

V. Results and Discussion

The study yields several key outcomes, which are given as follows:

A. Map-Matching

In this study, the Map-Matching process is implemented using the KD-Tree method. It operates by identifying the nearest edges from the 'maid' coordinates within the road network.

As shown in Table IV, the latitude and longitude coordinates are transformed into adjusted latitude and adjusted longitude to align with the nearest road network coordinates.

B. Compression

The efficiency of various GPS data compression methods in representing original movement has been analyzed. Four compression methods tested include Douglas-Peucker [15], Dead Reckoning [16], Immobility Filter, and a combination of Dead Reckoning and Immobility Filter. To understand the quality of the compressed data representation, each compressed dataset was compared to the raw data (ground truth) using quality metrics such as Precision (4), Recall (5), and F1-Score (6).

$$Precision = \frac{TP}{TP + FP} \tag{4}$$

$$Recall = \frac{TP}{TP + FN} \tag{5}$$

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{6}$$

Table V compares four data compression methods, with Immobility Filtering showing the highest Recall and F1-Score. This indicates a closer match to raw data due to the method's approach of selectively removing data only when the 'maid' remains immobile for a defined period. By retaining moving data, Immobility Filtering aims to enhance the accuracy of information available for path reconstruction.

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TABLE IV
DURING THE MAP MATCHING STEP, THE 'MAID' COORDINATE ARE SHIFTED TO THE NEARST ROAD NETWORK.

maid	latitude	longitude	timestamp	adjusted lat	adjusted lon
2390304d-c79a-4418-b435-12124cdbb782	-7.802600	110.409170	01/04/22 23.46	-7.802547	110.408961
2390304d-c79a-4418-b435-12124cdbb782	-7.802596	110.409180	01/04/22 23.46	-7.802547	110.408961
2390304d-c79a-4418-b435-12124cdbb782	-7.806945	110.411064	02/04/22 00.07	-7.807085	110.411341
2390304d-c79a-4418-b435-12124cdbb782	-7.782212	110.462814	02/04/22 00.46	-7.781726	110.462155

TABLE V
COMPARISON OF COMPRESSION METHODS

Compression Method	Precision	Recall	F1-Score
Ramer Douglas Peucker with ϵ = 0.0001	1.0	0.1312661	0.23206940
Dead Reconing	1.0	0.0981937	0.17882772
Dead Reconing + Immobility Filter	1.0	0.0952547	0.17394082
Immobility Filter	1.0	0.2385217	0.38517168

C. Path Recognition

Fig. 4 illustrates that the connections between coordinate points are relatively distant, resulting in connecting lines that fail to accurately represent the road network pattern. This study leverages a road network graph in the proposed path recognition method, which involves adding new coordinate points to fill in the gaps between two original coordinate points. These new coordinates are added based on the shortest path derived from the A* algorithm.

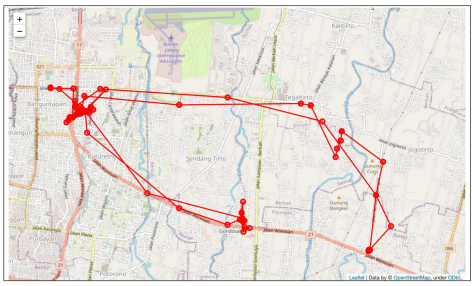

Fig. 4. Data Visualization After Map-Matching.

Fig. 5 provides an illustration of the road network graph, where each newly added data point represents a vertex.

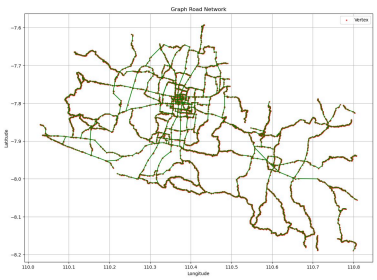

Fig. 5. Road Network Graph Visualization.

The new coordinates added to refine the shape of the trajectory initially lack timestamps, necessitating the needs to estimate the time of these new coordinates using a method known as time interpolation. Time interpolation is derived from the calculations where the time difference between two original coordinate points are divided by the number of new coordinate points spawned in between those two original coordinates. Fig. 6 below illustrates the concept of time interpolation.

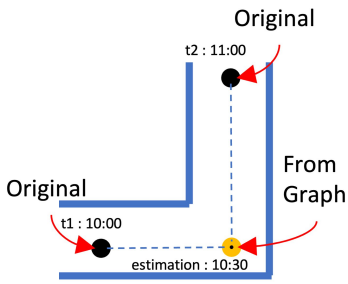

Fig. 6. Time Interpolation.

Upon completion of all the processes, the result of the path reconstruction is as depicted in Fig. 7. The difference is evident when comparing Fig. 4, which shows the state before path reconstruction, and Fig. 7, which illustrates the paths after the reconstruction process has been applied.

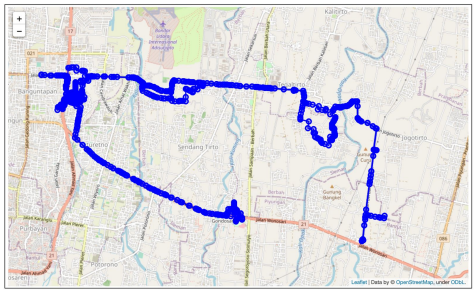

Fig. 7. Data Visualization After Path Reconstruction.

VI. CONCLUSIONS

This paper introduces a new technique for reconstructing paths from low-sampling-rate data. The primary challenge in path reconstruction from this data type is the irregular distribution of coordinate points and the considerable amount of noise that needs to be managed in the early stages of processing. The cornerstone of this road network path formation is the employment of a road network graph, which favors the addition of new coordinates based on the locations of vertices that align with the network's layout.This research has succeeded in creating path reconstruction which can be useful for many future studies. However, there are some limitations to the proposed method, including the inability to differentiate one-way streets due to data limitations and insufficient road network map information. In general, the method proposed exhibits significant capabilities and potential for producing road network maps using low-sampling-rate GPS trajectory data.

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